

Yogendra A

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Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL, C
Backend & APIs: FastAPI, Flask, REST APIs, AsyncIO, WebSockets, Server-Sent Events (SSE), MVC
Databases: MySQL, MongoDB, Firebase Firestore, Schema Design, SQL Queries, Triggers, Stored Procedures
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, Concurrency, API Design, Unit Testing, Debugging, Git, GitHub
Cloud & DevOps: Docker, Linux, GitHub Actions, CI/CD, AWS, GCP
Frontend: React.js, Vite, Tkinter
AI/ML: PyTorch, Hugging Face, RAG, FAISS, Whisper, CNN-LSTM

Work Experience

L&T Technology Services (LTTS) — Audio Intelligence Feb 2026 – May 2026

Intern

- Engineered a canonical **14-task Python framework** with typed result contracts, bounded concurrent execution, lazy thread-safe model caching, and structured task-failure isolation across multiple processing paths.
- Implemented bounded concurrency using **ThreadPoolExecutor**, centralized task registration and cross-path coverage, and improved consistency across sequential, parallel, and file-processing execution paths.
- Strengthened reliability through **13/13 regression tests**, fault-injection scenarios, graceful handling of missing model dependencies, and deterministic fallback behavior for processing failures.
- Debugged and improved heterogeneous Python/ML components across Windows and Linux environments, including model-loading, caching, task integration, and runtime reliability issues.

Projects

VoiceRAG — Voice-Driven RAG Knowledge Assistant Jan 2025 – Mar 2025

- Built a **FastAPI backend** exposing document ingestion, semantic chunking, and FAISS-based vector search endpoints, integrated with a React/Vite frontend.
- Implemented **Server-Sent Events (SSE)** for streaming responses and integrated Whisper transcription as an upstream service for voice-driven document queries.

Pranayama Trainer — Breathing & Posture Analysis Dec 2024 – Mar 2025

- Built a Python **AsyncIO + WebSockets backend** handling concurrent webcam and audio streams, with session data persisted through Firebase Firestore.
- Designed the API and data layer connecting a React frontend to the backend inference pipeline; integrated a CNN-LSTM classifier achieving **78.4% test accuracy** during internal validation.

Recruitment Management System 2024

- Designed a normalized **5-table MySQL schema** with foreign-key constraints and application-state validation, implementing triggers for business-rule enforcement and a stored procedure for cascading job deletion.
- Built the Python application layer using Tkinter and mysql-connector with separate recruiter and candidate workflows for job posting, applications, and status tracking.

Education

PES University, Bengaluru Sep 2022 – Aug 2026

B.Tech in Computer Science & Engineering — CGPA: 7.17 / 10

- Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Machine Learning

Certifications

- Goldman Sachs Software Engineering Virtual Experience** — IT security simulation; identified outdated password hashing via Hashcat and proposed algorithm migration and password policy improvements.
- LangChain Deep Agents Certificate** — Mar 2026; agentic AI workflows using LangGraph, tool calling, and multi-step reasoning patterns.
- Webflow 101 Certification** — Visual web design, component-based layouts, animations, and responsive UI.